

CM0081 Formal Languages and Automata

§ 3.4 Algebraic Laws for Regular Expressions

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Preliminaries

Textbook

John E. Hopcroft, Rajeev Motwani and Jefferey D. Ullman (2007). Introduction to Automata Theory, Languages, and Computation.

Conventions

- The number and page numbers assigned to chapters, examples, exercises, figures, quotes, sections and theorems on these slides correspond to the numbers assigned in the textbook.
- The natural numbers include the zero, that is, $\mathbb{N} = \{0, 1, 2, \dots\}$.
- The power set of a set A , that is, the set of its subsets, is denoted by $\mathcal{P}A$.

Algebraic Laws for Regular Expressions

Definition

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Let M , N and P be regular expression variables.

Sugar syntax

$$M^+ := MM^*,$$

$$M? := \varepsilon + M.$$

Algebraic Laws for Regular Expressions

Some laws for union

$$(M + N) + P = M + (N + P)$$

(associativity)

$$M + \emptyset = \emptyset + M = M$$

(identity)

$$M + N = N + M$$

(commutativity)

$$M + M = M$$

(idempotence)

Remark

There is no inverse for union.

Algebraic Laws for Regular Expressions

Some laws for concatenation

$$(MN)P = M(NP)$$

(associativity)

$$M\varepsilon = \varepsilon M = M$$

(identity)

$$MN \neq NM$$

(non-commutativity)

$$M\emptyset = \emptyset M = \emptyset$$

($\emptyset$ is the annihilator for concatenation)

Remark

There is no inverse for concatenation.

Algebraic Laws for Regular Expressions

Some laws for union and concatenation

$$M(N + P) = MN + MP \quad \text{(distributive)}$$

$$(M + N)P = MP + NP \quad \text{(distributive)}$$

Algebraic Laws for Regular Expressions

Some laws for closure

$$(M^*)^* = M^*$$

(idempotence)

$$\emptyset^* = \varepsilon$$

$$\varepsilon^* = \varepsilon$$

$$(\varepsilon + M)^* = M^*$$

$$M^* = M^+ + \varepsilon$$

Algebraic Laws for Regular Expressions

Remark

A complete set of axioms for the regular expressions is indicated in (Kozen 2012, Lecture 9).

Simplification of Regular Expressions

Example

$$\begin{aligned}0 + (\varepsilon + 1)\underline{(\varepsilon + 1)^*}0 &= 0 + \underline{(\varepsilon + 1)1^*}0 \\&= 0 + (\underline{\varepsilon 1^*} + 11^*)0 \\&= 0 + (1^* + \underline{11^*})0 \\&= 0 + (\underline{1^*} + 1^+)0 \\&= \underline{0 + 1^*}0 \\&= 1^*0\end{aligned}$$

$$((\varepsilon + M)^* = M^*)$$

(distributive)

(identity)

(def. L^+)

(equivalence)

(equivalence)

Discovering Laws for Regular Expressions

Method

Let E and F be two regular expressions with the same set of variables $\{M_1, \dots, M_n\}$.

To test if $E = F$:

1. Convert E and F to **concrete** regular expressions C and D , replacing each M_i by a different symbol a_i , for $i = 1, 2, \dots, n$.
2. Test whether $L(C) = L(D)$. If so, then $E = F$, and if not $E \neq F$.

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Observation

We are proving by example!

Discovering Laws for Regular Expressions

Example

Prove or disprove that $M^* = M^*M^*$.

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We replace the variable M by the concrete regular expression a .

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Because $L(a^*) = L(a^*a^*)$ we conclude that

$$M^* = M^*M^*.$$



Discovering Laws for Regular Expressions

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Refutation

We replace the variables M and N by the concrete regular expressions a and b respectively.

$$a + ba \stackrel{?}{=} (a + b)a.$$

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We replace the variables M and N by the concrete regular expressions a and b respectively.

$$a + ba \stackrel{?}{=} (a + b)a.$$

Because $aa \notin L(a + ba)$ and $aa \in L((a + b)a)$ then

$$L(a + ba) \neq L((a + b)a).$$

So, we can conclude

$$M + NM \neq (M + N)M.$$



Discovering Laws for Regular Expressions

Example (Exercise 3.4.2.d)

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Discovering Laws for Regular Expressions

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Refutation

We replace the variables M and N by the concrete regular expressions a and b respectively.

$$(a + b)^*b \stackrel{?}{=} (a^*b)^*.$$

Since $\varepsilon \notin (a + b)^*b$ and $\varepsilon \in (a^*b)^*$ then

$$(M + N)^*N \neq (M^*N)^*.$$



Discovering Laws for Regular Expressions

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1. Add $\cap$ to the regular expression operators.
2. Test if $M \cap N \cap P = M \cap N$.
3. From $M = a$, $N = b$ and $P = c$ and since

$$\{a\} \cap \{b\} \cap \{c\} = \emptyset = \{a\} \cap \{b\},$$

we should conclude that the “property” is true.

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we should conclude that the “property” is true.

4. But, the “property” is false. For example, if $M = N = a$ and $P = \emptyset$ then

$$M \cap N \cap P \neq M \cap N.$$

5. Therefore, the test is not valid!

References

- John E. Hopcroft, Rajeev Motwani and Jefferey D. Ullman [1979] (2007). Introduction to Automata Theory, Languages, and Computation. 3rd ed. Pearson Education (cit. on p. 2).
- Dexter C. Kozen [1997] (2012). Automata and Computability. Third printing. Undergraduate Texts in Computer Science. Springer. DOI: [10.1007/978-1-4612-1844-9](https://doi.org/10.1007/978-1-4612-1844-9) (cit. on p. 10).